



Forces and Newton's Laws of Motion

• Mark Scheme

Mechanics Chapter 3: Forces and Newton's Laws of Motion. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Types of force and force diagrams

- 1 Weight W (or mg) downwards $B1$; normal reaction R upwards $B1$.
- 2 Weight down and reaction up $B1$; tension forwards and friction backwards $B1$.
- 3 (a) $R = 3 \times 9.8 = 29.4 \text{ N } B1$. (b) $R + 10 = 29.4 \text{ M1}$; $R = 19.4 \text{ N } A1$.
- 4 Resultant: $3 \text{ N west and } 12 \text{ N north } M1$; magnitude $\sqrt{153} = 12.4 \text{ N } A1$; bearing 346° A1 .
- 5(a) Weight 39.2 N down , R up, tension at 30° , friction opposite to the horizontal component $B2$.
- 5(b) $R(\uparrow): R + 20\sin 30^\circ = 39.2 \text{ M1}$; $R = 29.2 \text{ N } A1$.
- 5(c) $F = 20\cos 30^\circ \text{ M1}$; $= 17.3 \text{ N } A1$.
- 5(d) The tension has an upward component, which supports part of the weight $B1$.

Section B – Newton's first and second laws

- 6 $12 = 3a \text{ M1}$; $a = 4 \text{ ms}^{-2} \text{ A1}$.
- 7 $30 - 10 = 5a \text{ M1}$; $a = 4 \text{ ms}^{-2} \text{ A1}$; $v = 12 \text{ ms}^{-1} \text{ A1}$.
- 8 $3000 - 600 = 1200a \text{ M1}$; $a = 2 \text{ ms}^{-2} \text{ A1}$; $t = 10 \text{ s } A1$.
- 9 $R(\rightarrow): 20\cos 30^\circ = 2a \text{ M1}$; $a = 8.66 \text{ ms}^{-2} \text{ A1}$; $R(\uparrow): R + 20\sin 30^\circ = 19.6 \text{ M1}$; $R = 9.6 \text{ N } A1$.
- 10(a) Whole system: $T - 580g = 580 \times 1.2 \text{ M1 } A1$; $T = 6380 \text{ N } A1$.
- 10(b) Man: $R - 80g = 80 \times 1.2 \text{ M1}$; $R = 880 \text{ N } A1$.
- 10(c) $R - 80g = 80 \times (-2) \text{ M1}$; $R = 624 \text{ N } A1$.
- 10(d) The resultant force is zero, so the tension equals the total weight (and R equals the man's weight) $B1$.

Section C – Newton's third law and connected particles

- 11 A third-law pair acts on *different* bodies and is the same type of force $B1$; both of these forces act on the book (the pair to the weight is the book's pull on the Earth) $B1$.
- 12 $2100 = 1400a \text{ M1}$; $a = 1.5 \text{ ms}^{-2} \text{ A1}$; trailer: $T = 400 \times 1.5 = 600 \text{ N } A1$.
- 13 $15 = 5a \Rightarrow a = 3 \text{ ms}^{-2} \text{ M1 } A1$; $B: T = 2 \times 3 = 6 \text{ N } A1$.
- 14 $3600 - 600 = 1500a \text{ M1}$; $a = 2 \text{ ms}^{-2} \text{ A1}$; trailer: $T - 200 = 300 \times 2 \text{ M1}$; $T = 800 \text{ N } A1$.
- 15(a) $6000 - 1500 = 3000a \text{ M1}$; $a = 1.5 \text{ ms}^{-2} \text{ A1}$; trailer: $T - 500 = 1000 \times 1.5 \text{ M1}$; $T = 2000 \text{ N } A1$.
- 15(b) $-(2500 + 1500) = 3000a \text{ M1}$; deceleration $\frac{4}{3} = 1.33 \text{ ms}^{-2} \text{ A1}$; trailer: $-500 - T' = 1000 \times (-\frac{4}{3})$, where T' is a thrust towards the back $M1$; thrust of $833 \text{ N } A1$.

Section D – Pulleys

- 16 $5g - 3g = 8a \text{ M1}$; $a = 2.45 \text{ ms}^{-2} \text{ A1}$.
- 17 $6g - T = 6a$, $T - 4g = 4a \text{ M1}$; $a = 1.96 \text{ ms}^{-2} \text{ A1}$; $T = 47.0 \text{ N } A1$.
- 18 $1g = 5a \Rightarrow a = 1.96 \text{ ms}^{-2} \text{ M1 } A1$; $T = 4 \times 1.96 = 7.84 \text{ N } A1$.



19 $a = 2.45 \text{ ms}^{-2}$ *B1*; $v^2 = 2(2.45)(1.5)$ *M1*; $v = 2.71 \text{ ms}^{-1}$ *A1*; $t = 1.11 \text{ s}$ *A1*.

20(a) *A*: $T = 2a$ *M1*; *B*: $3g - T = 3a$ *M1*; $a = 5.88 \text{ ms}^{-2}$ *A1*; $T = 11.8 \text{ N}$ *A1*.

20(b) $v^2 = 2(5.88)(1.2)$ *M1*; $v = 3.76 \text{ ms}^{-1}$ *A1*.

20(c) First stage: $t_1 = \sqrt{\frac{2 \times 1.2}{5.88}} = 0.639 \text{ s}$ *M1*; then *A* moves the remaining 0.8 m at constant speed: $t_2 = \frac{0.8}{3.757} = 0.213 \text{ s}$ *M1*; total 0.852 s *A1*.

Roar-level challenge

21 $4g - T = 4(2.8)$ *M1*; $T - mg = 2.8m$ *M1*; $T = 39.2 - 11.2 = 28 \text{ N}$ *A1*; $28 = 12.6m$ *M1*; $m = \frac{20}{9} = 2.22$ *A1*.